Spatial proximity & ligand–receptor analysis

Nandhu_version_final cell types · 49 ROIs (23 non-relapse, 26 relapse) · 233,949 cells · tumour = NPM1_blasts (43,493 cells) · 2026-11-11

What was done

• **Tumour.** Every cell labelled NPM1_blasts in Nandhu_version_final. This includes all v2 Tumor and Tumor/Erythroid cells.

• **Distance.** The nucleus-centre distance from each cell to its nearest NPM1 blast in the same ROI, at 0.325 µm per pixel. <15 µm means the cells are touching.

• **Part 1, proximity screen** (June method). The % of each cell type within 32.5 µm of a blast.

• **Part 2, distance zones** (June method). The cell-type make-up of non-tumour cells in 5 zones: <15, 15–30, 30–60, 60–100 and 100+ µm.

• **Part 3, ligand–receptor** (July 3 method). For every blast–immune cell pair closer than 100 µm, the spatial interaction score is SIS = mean(max(z_{ligand},0) × max($z_{receptor}$,0)). The four pairs are PD-L1→PD-1, Galectin-9→TIM-3, Galectin-9→CD44 and HLA-DR→LAG-3. Receivers are scored as all immune cells and as T cells only.

• **Parts 4–5, pathway modules** (July 3 method). 12 marker modules are scored per immune cell as the mean per-machine z of their markers. They are compared by distance zone (Contact <15, Near 15–30, Mid 30–50, Far 50–100 µm) for all immune cells together and then separately for T cells, Myeloid and Plasma cells.

• **Statistics.** One ROI is one replicate. Two-sided Mann–Whitney U, with Benjamini–Hochberg correction inside each family (within zone, or within lineage × zone). Unassigned/Other and Artifact are not cell types, so they are left out of every composition.

Key findings

• **Relapse marrow is dominated by blasts, so everything sits closer to them.** NPM1 blasts are 27.9% of annotated cells in relapse ROIs versus 10.3% in non-relapse ROIs (BH p = 0.003). As a result, every cell type is 1.5–2× more often within 32.5 µm of a blast in relapse (all 14 types BH < 0.05; Fig 1). This is mostly a consequence of tumour burden. It is not on its own evidence of targeted recruitment, which is why the zone analysis below compares *composition* at a fixed distance.

• **Blood vessels mark the non-relapse niche (the strongest result).** Endothelial cells are about 3× more common around blasts in non-relapse at **every** distance, from touching (8.6% vs 2.9%) to far (8.7% vs 1.7%), with BH p ≤ 0.004 in all 5 zones (Figs 2–3). Non-relapse blasts sit in vascular-rich marrow. Relapse blasts sit in vessel-poor tissue.

• **The v2 dendritic-cell population (now 'Immature myeloid CD11c+') touches blasts only in non-relapse.** It is 3.6% of cells touching blasts in non-relapse versus 0.16% in relapse (BH 0.034). It is also enriched at 60–100 µm (BH 0.041). This reproduces the June dendritic-cell finding with the corrected labels.

• **T cells are packed against the blasts in relapse,** but they are not proliferating. In the touching zone, CD8+ T are 7.9% vs 3.0% (BH 0.011) and CD4+ T are 5.3% vs 2.8% (BH 0.022). The Ki67+CD147+CD45RA+ Leiden population is 3.1% vs 0% (BH 0.007).

• **Immune proliferation is switched off at the blast surface in relapse (ROI level).** This replicates the July 3 headline. For all immune cells, the Proliferation module (Ki-67/PCNA/RRM2/c-MYC) at contact is lower in relapse (BH 0.040; Figs 8, 10). By lineage: T cells at contact BH 0.013 and near BH 0.049, Myeloid at contact BH 0.044 (Figs 11–12).

• **Wnt/β-catenin is raised in myeloid cells touching blasts in relapse** (BH 0.044), and it trends up across all zones. **Type-I IFN (IRF7/IFN-γ) is lower in T cells at contact in relapse** (BH 0.023).

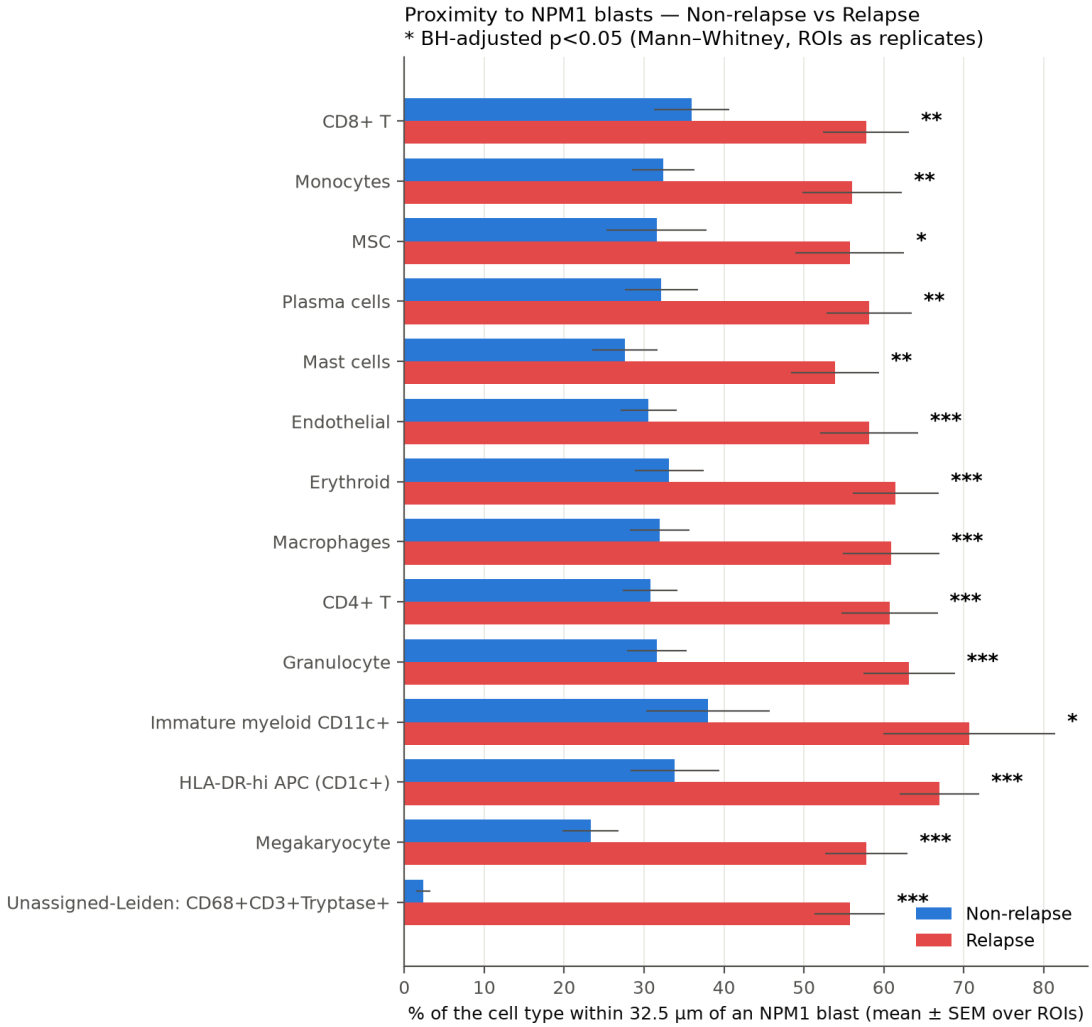
• **Patient-level check (Part 6).** Several ROIs come from the same patient, so the key results were re-tested with one value per patient (6 non-relapse vs 5 relapse). **The endothelial result holds**: BH 0.043 in 3 of 5 zones, raw p ≤ 0.017 in all 5. **The immune-state results (proliferation, IFN, Wnt) and the CD11c+/T-cell contact differences point the same way but are not significant per patient** (p 0.15–0.55). Treat them as ROI-level findings that need

more patients.

• **No ligand–receptor pair separates the groups** (all BH > 0.35; Figs 5–7). PD-L1→PD-1 on T cells is not higher in relapse at contact (0.14 vs 0.17).

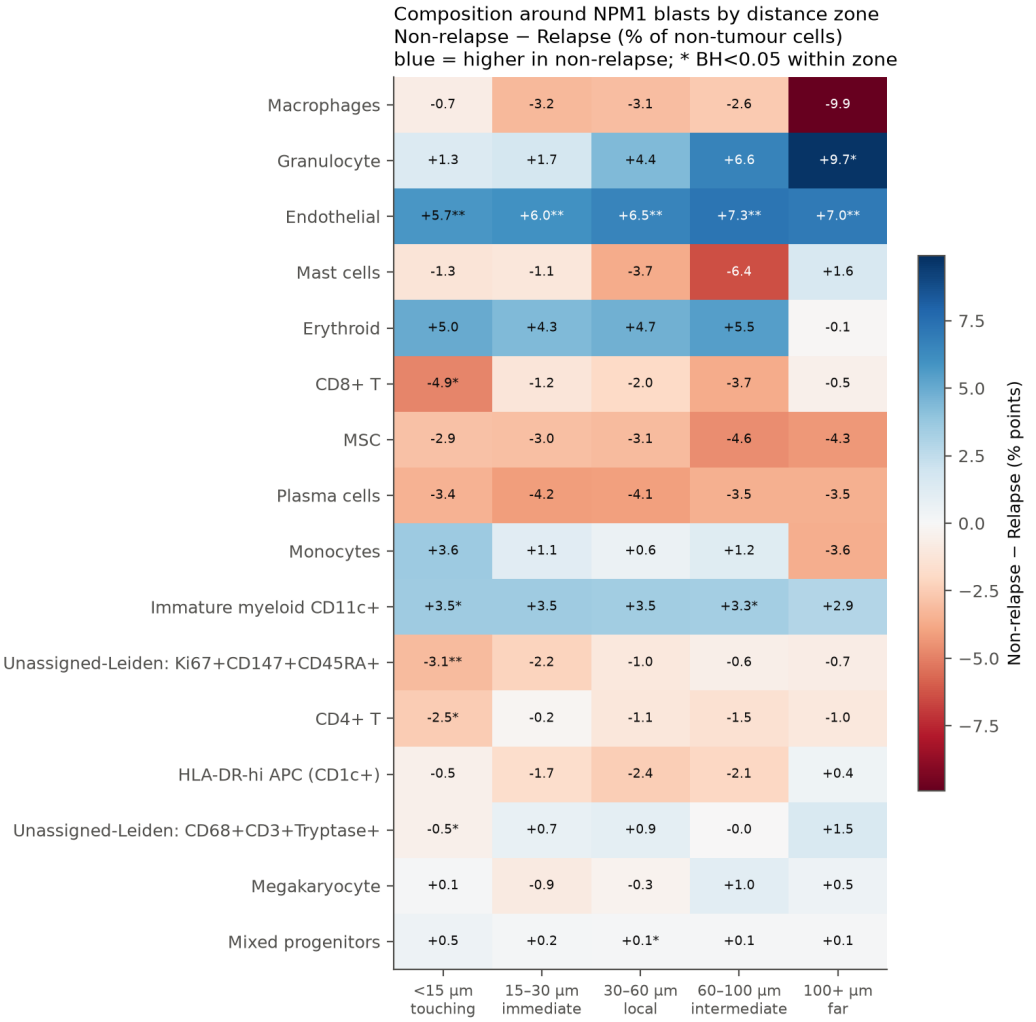
• **Not reproduced with these labels:** the July findings of lower T-cell Bcl-2 at contact (now raw p 0.031, BH 0.10) and lower myeloid HLA-DR (p 0.25). The far-zone macrophage excess in relapse is also no longer significant (p 0.22). Read these as trends only.

Part 1 — proximity screen



Cell type	Non-relapse %	Relapse %	p	BH p
Megakaryocyte	23.3	57.8	<0.001	<0.001
HLA-DR-hi APC (CD1c+)	33.9	67.0	<0.001	<0.001
CD4+ T	30.8	60.7	<0.001	<0.001
Granulocyte	31.6	63.2	<0.001	<0.001
Erythroid	33.1	61.5	<0.001	<0.001
Endothelial	30.6	58.2	<0.001	<0.001
Unassigned-Leiden: CD68+CD3+Tryptase+	2.4	55.7	<0.001	<0.001
Macrophages	32.0	60.9	<0.001	<0.001
Mast cells	27.6	53.9	0.001	0.002
Plasma cells	32.1	58.2	0.002	0.002
CD8+ T	36.0	57.8	0.004	0.005
Monocytes	32.4	56.0	0.004	0.005
MSC	31.6	55.7	0.024	0.025
Immature myeloid CD11c+	38.0	70.7	0.034	0.034

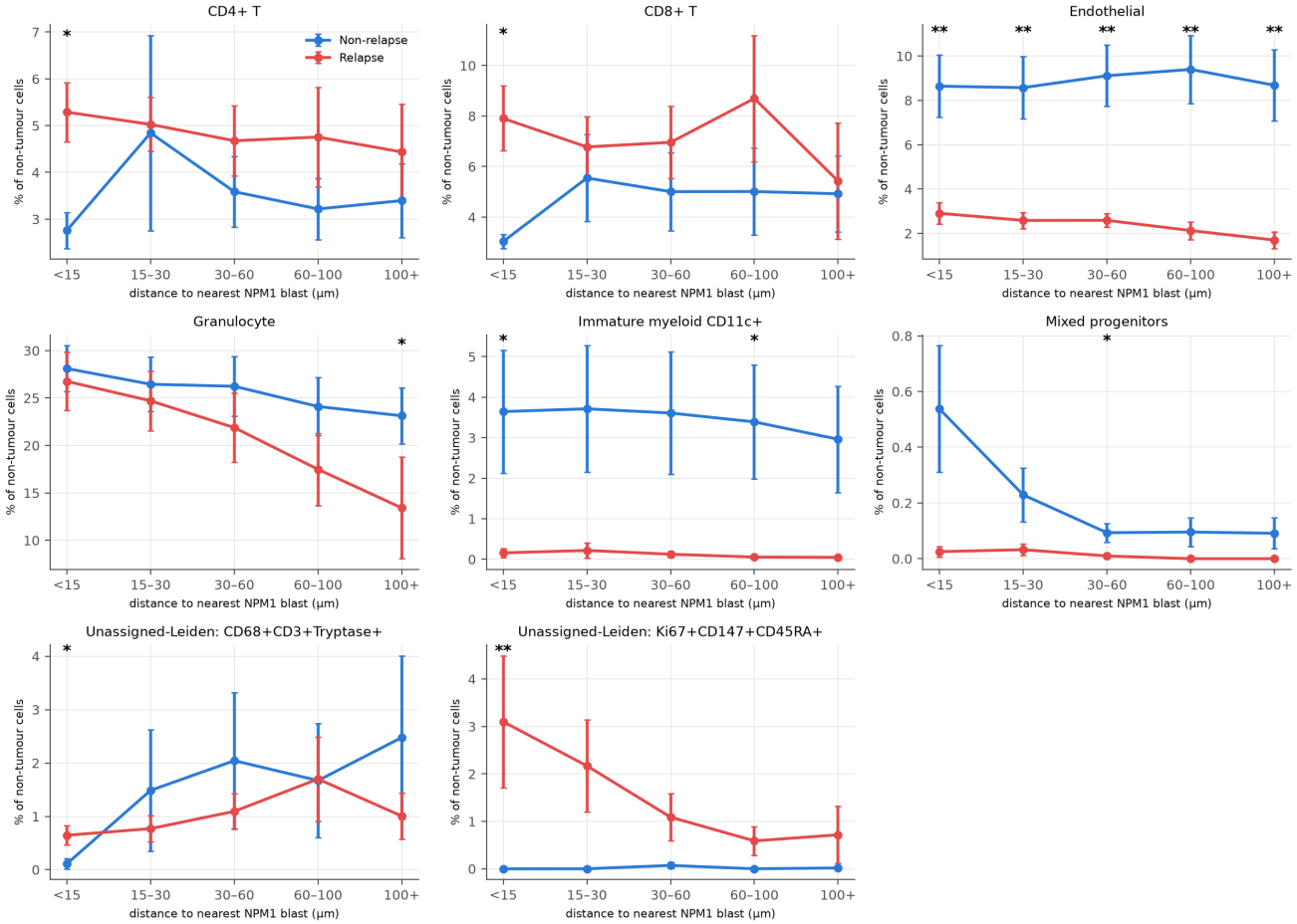
Part 2 — composition by distance zone



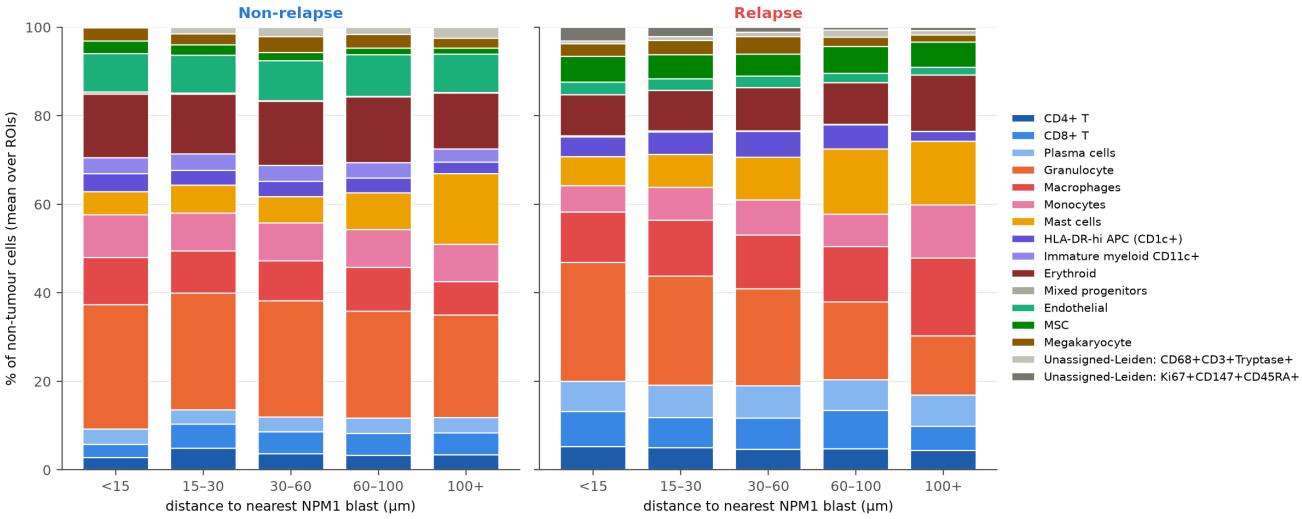
Significant zone differences (BH < 0.05 within zone):

Zone	Cell type	Non-relapse %	Relapse %	BH p	Higher in
<15 µm touching	Endothelial	8.64	2.91	0.004	Non-relapse
<15 µm touching	Unassigned-Leiden: Ki67+CD147+CD45RA+	0.00	3.10	0.007	Relapse
<15 µm touching	CD8+ T	3.03	7.91	0.011	Relapse
<15 µm touching	CD4+ T	2.76	5.29	0.022	Relapse
<15 µm touching	Immature myeloid CD11c+	3.65	0.16	0.034	Non-relapse
<15 µm touching	Unassigned-Leiden: CD68+CD3+Tryptase+	0.11	0.65	0.034	Relapse
15–30 µm immediate	Endothelial	8.57	2.58	0.007	Non-relapse
30–60 µm local	Endothelial	9.11	2.59	0.001	Non-relapse
30–60 µm local	Mixed progenitors	0.09	0.01	0.026	Non-relapse
60–100 µm intermediate	Endothelial	9.40	2.13	0.001	Non-relapse
60–100 µm intermediate	Immature myeloid CD11c+	3.39	0.06	0.041	Non-relapse
100+ µm far	Endothelial	8.68	1.70	0.002	Non-relapse
100+ µm far	Granulocyte	23.13	13.42	0.035	Non-relapse

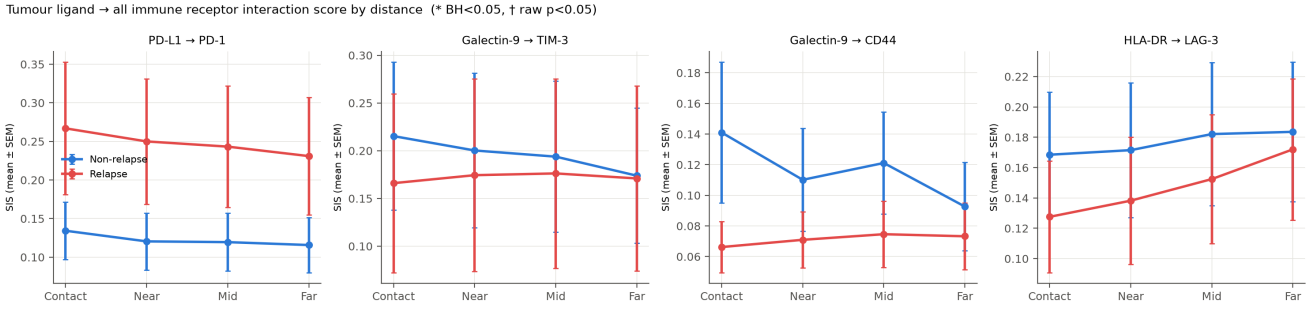
Cell types that differ in at least one zone (mean ± SEM; * BH<0.05)

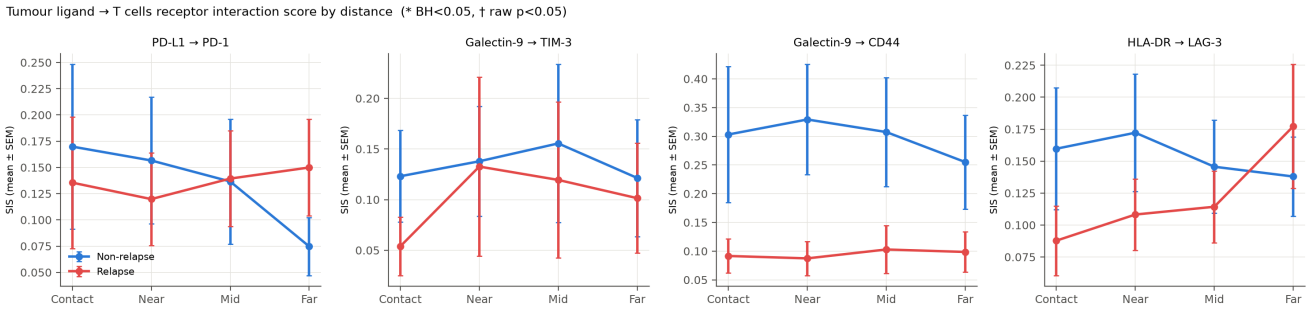


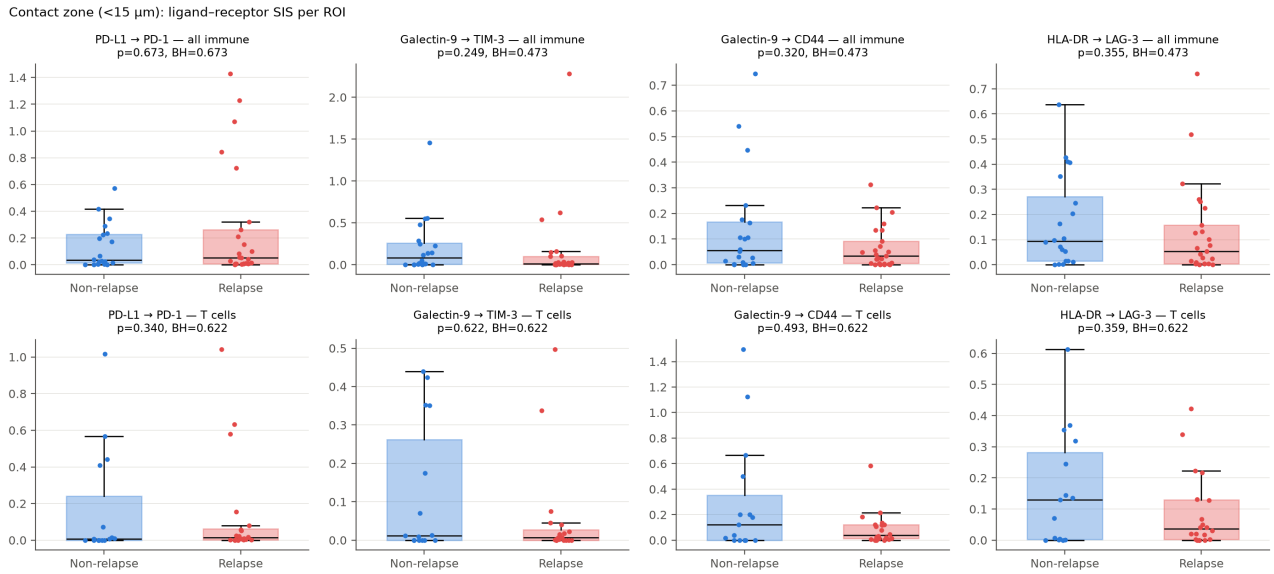
Non-tumour composition by distance from NPM1 blasts



Part 3 — ligand–receptor interaction scores



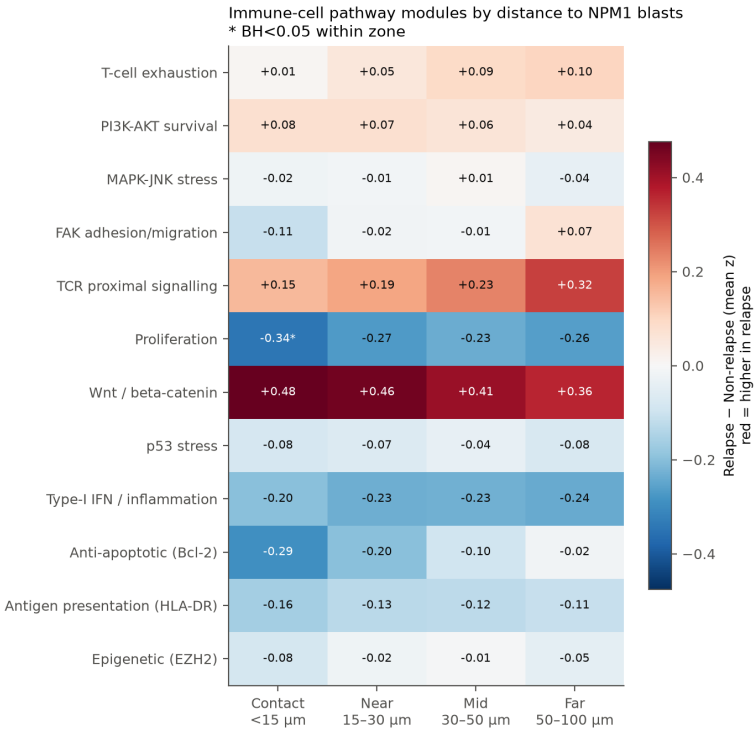


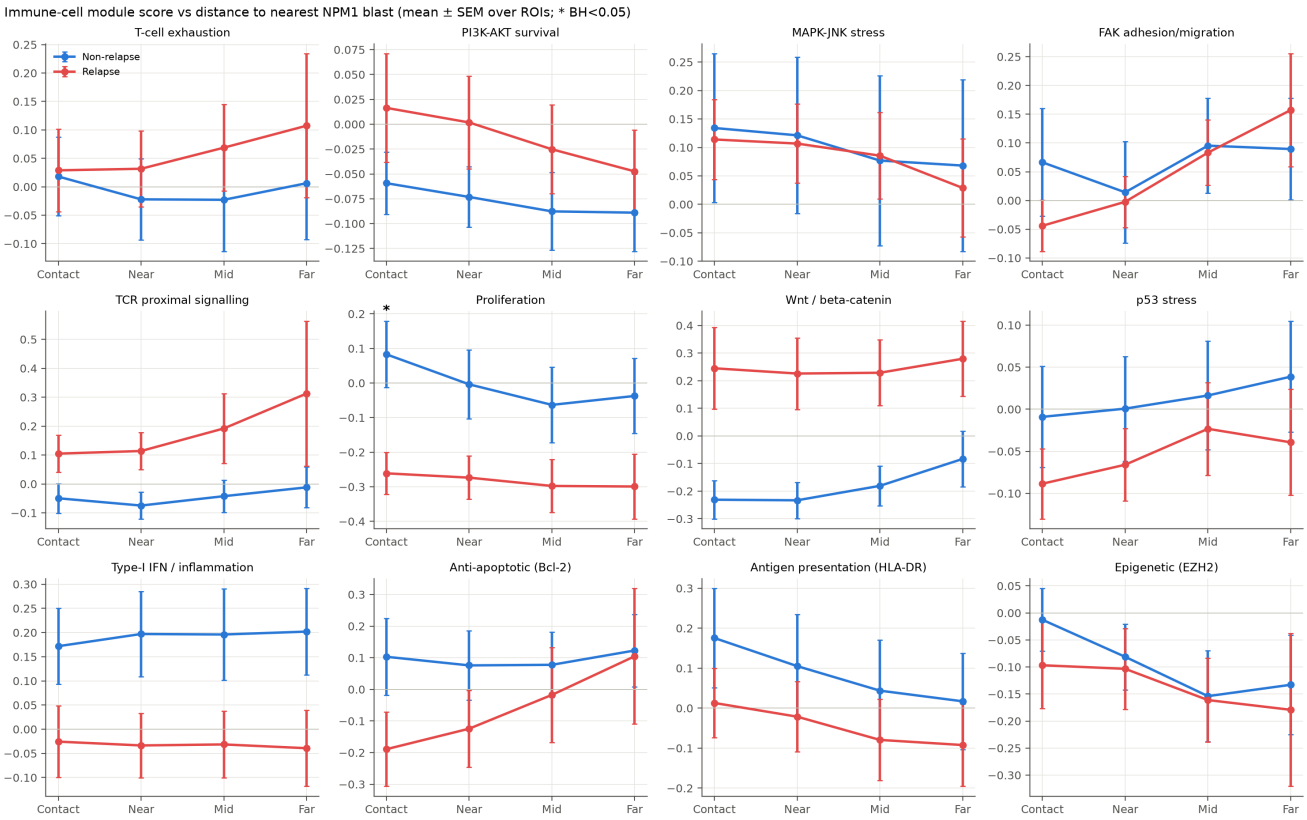


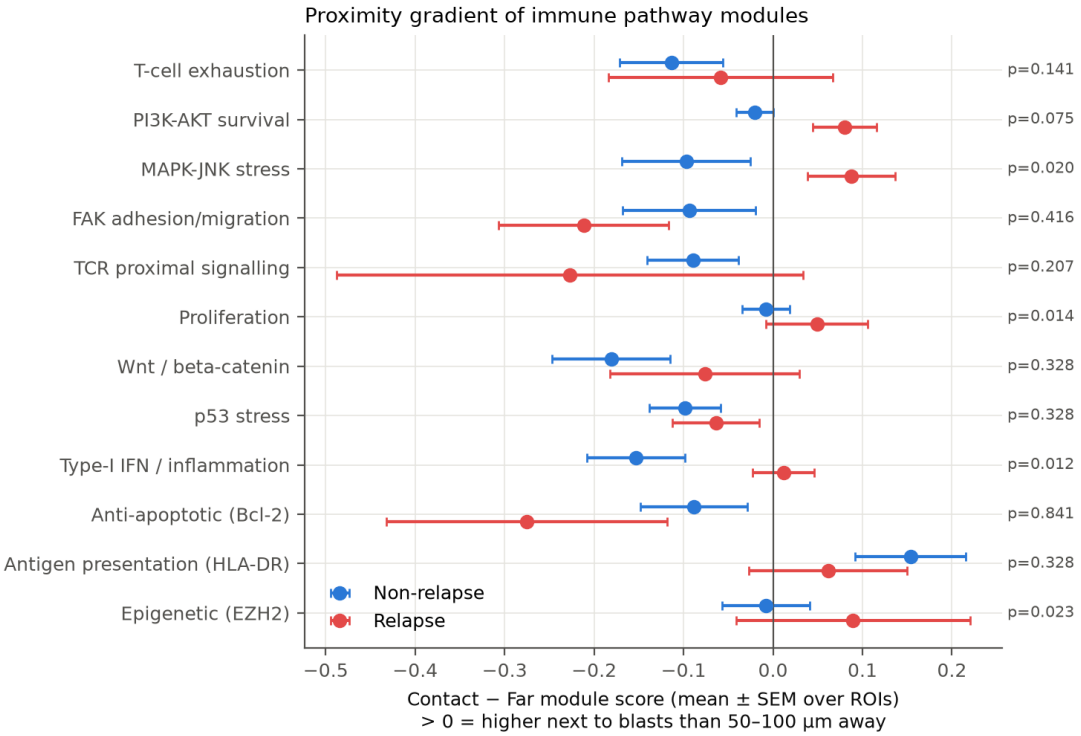
Contact zone (<15 µm):

Receiver	Pair	Non-relapse SIS	Relapse SIS	p	BH p
T cells	Galectin-9 → CD44	0.303	0.092	0.493	0.622
T cells	Galectin-9 → TIM-3	0.123	0.054	0.622	0.622
T cells	HLA-DR → LAG-3	0.160	0.088	0.359	0.622
T cells	PD-L1 → PD-1	0.170	0.135	0.340	0.622
all immune	Galectin-9 → CD44	0.141	0.066	0.320	0.473
all immune	Galectin-9 → TIM-3	0.215	0.166	0.249	0.473
all immune	HLA-DR → LAG-3	0.169	0.128	0.355	0.473
all immune	PD-L1 → PD-1	0.134	0.267	0.673	0.673

Part 4 — immune pathway modules vs distance





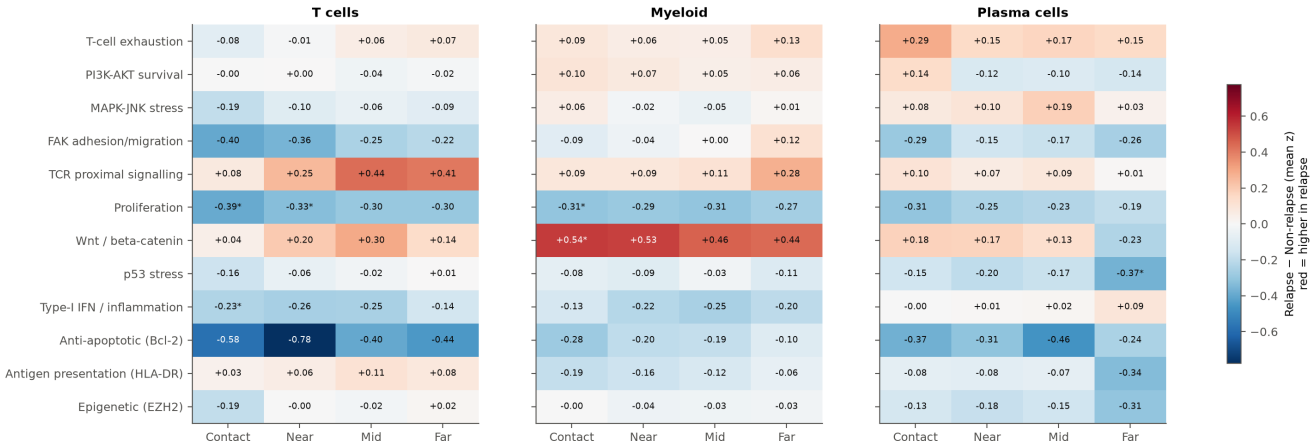


Contact – Far gradient. Nothing survives correction. The strongest raw trends (Type-I IFN, Proliferation, MAPK-JNK and EZH2) all rise towards the blasts in relapse and fall in non-relapse.

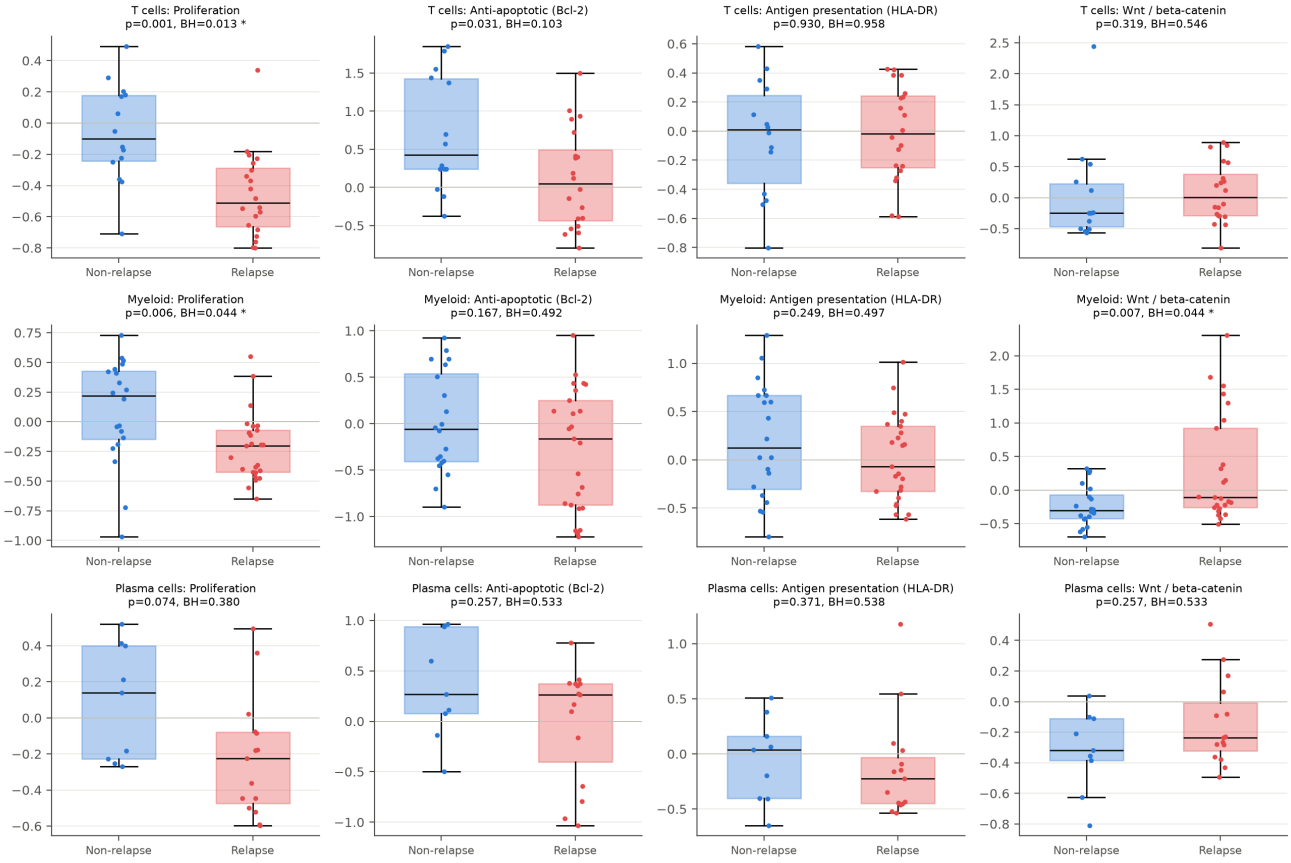
Module	Gradient NR	Gradient R	p	BH p
Type-I IFN / inflammation	-0.153	0.012	0.012	0.069
Proliferation	-0.008	0.049	0.014	0.069
MAPK-JNK stress	-0.097	0.088	0.020	0.069
Epigenetic (EZH2)	-0.008	0.090	0.023	0.069
PI3K-AKT survival	-0.020	0.080	0.075	0.180
T-cell exhaustion	-0.114	-0.059	0.141	0.281
TCR proximal signalling	-0.090	-0.227	0.207	0.355
Antigen presentation (HLA-DR)	0.154	0.062	0.328	0.394
Wnt / beta-catenin	-0.181	-0.076	0.328	0.394
p53 stress	-0.099	-0.064	0.328	0.394
FAK adhesion/migration	-0.094	-0.212	0.416	0.454
Anti-apoptotic (Bcl-2)	-0.088	-0.275	0.841	0.841

Part 5 — lineage-resolved modules

Lineage-resolved pathway modules by distance zone (* BH<0.05 within lineage × zone)



Contact zone (<15 µm): key modules per immune lineage, one dot per ROI



Lineage	Zone	Module	Non-relapse	Relapse	BH p
Myeloid	Contact <15 µm	Proliferation	0.09	-0.22	0.044
Myeloid	Contact <15 µm	Wnt / beta-catenin	-0.24	0.30	0.044
Plasma cells	Far 50–100 µm	p53 stress	0.16	-0.21	0.045
T cells	Contact <15 µm	Proliferation	-0.06	-0.46	0.013
T cells	Contact <15 µm	Type-I IFN / inflammation	-0.09	-0.32	0.023
T cells	Near 15–30 µm	Proliferation	-0.12	-0.45	0.049

Part 6 — patient-level check of the key results

Per-ROI values are averaged within each patient, then compared 6 vs 5 (Mann–Whitney, exact; the smallest possible p is 0.0043). Contact-zone endpoints have 5 non-relapse patients because one patient has too few blasts for a contact zone.

Endpoint	NR mean	R mean	n pts NR/R	p (patient)	BH
Endothelial % — <15 µm touching	9.54	3.35	5/5	0.016	0.052
Endothelial % — 15–30 µm immediate	9.11	3.03	6/5	0.009	0.043
Endothelial % — 30–60 µm local	9.79	2.95	6/5	0.009	0.043
Endothelial % — 60–100 µm intermediate	10.46	2.43	6/5	0.017	0.052
Endothelial % — 100+ µm far	9.81	1.34	6/5	0.004	0.043
Immature myeloid CD11c+ % — <15 µm touching	5.48	0.34	5/5	0.151	0.323
CD8+ T % — <15 µm touching	3.42	6.97	5/5	0.310	0.422
CD4+ T % — <15 µm touching	2.83	5.01	5/5	0.151	0.323
All immune: Proliferation — Contact	-0.06	-0.18	5/5	0.548	0.548
T cells: Proliferation — Contact	-0.19	-0.32	5/5	0.548	0.548
T cells: Type-I IFN / inflammation — Contact	-0.09	-0.20	5/5	0.421	0.526
T cells: Anti-apoptotic (Bcl-2) — Contact	0.76	0.33	5/5	0.310	0.422
Myeloid: Proliferation — Contact	-0.05	-0.13	5/5	0.548	0.548
Myeloid: Wnt / beta-catenin — Contact	-0.29	0.17	5/5	0.310	0.422
Myeloid: Antigen presentation (HLA-DR) — Contact	0.35	-0.07	5/5	0.310	0.422

Biological interpretation

1. The non-relapse blast lives in a vascular, immune-active niche. Endothelium is consistently ~3× more abundant around non-relapse blasts at every distance, and the CD11c+ myeloid cells (the v2 'dendritic cells') make direct contact with blasts only in non-relapse. Together with proliferating immune cells at the contact zone, this describes an interface where antigen-presenting cells reach the blasts and immune cells are still expanding.

2. Relapse blasts may build an 'immune-cold shell' (ROI-level evidence). In relapse marrow the blast surface is crowded with CD4+ and CD8+ T cells, yet those T cells and the neighbouring myeloid cells are proliferation-arrested, T cells have lower type-I interferon signalling, and myeloid cells switch on the tolerogenic Wnt/β-catenin programme. The T cells are present but not expanding. This is a locally enforced suppression, strongest exactly at contact, not a lack of immune infiltration. These module differences are significant across ROIs but not yet across patients, so they need confirmation in more patients.

3. No single checkpoint explains it. None of PD-L1→PD-1, Galectin-9→TIM-3/CD44 or HLA-DR→LAG-3 differs between the groups. The relapse phenotype is a coordinated change in immune-cell *state* next to the blast, rather than one ligand–receptor axis.

Clinical interpretation

• **Possible spatial relapse biomarker.** Three features separate relapse from non-relapse in MRD-positive remission marrow while the blast count alone does not capture them: endothelial depletion around blasts (patient-level confirmed), loss of CD11c+ myeloid contact, and immune proliferation arrest at the blast surface. If they are validated in an independent cohort, a combined score could stratify relapse risk beyond MRD transcript levels.

• **Therapy rationale.** PD-1/PD-L1 blockade alone is unlikely to be enough. The data point to strategies that restore immune proliferation and priming (for example, combinations that relieve local suppression or target Wnt/β-catenin in myeloid cells), and possibly to the vascular niche as a factor in disease control.

• **Caveats.**

– 11 patients (6 non-relapse, 5 relapse) and 49 ROIs. ROI-level tests treat ROIs from the same patient as independent. Part 6 repeats the key endpoints per patient; only the endothelial result is confirmed there.

– Machine is confounded with group: non-relapse is 12 Machine A + 11 Machine B ROIs, relapse is 22 A + 4 B. Module markers are z-scored within machine to reduce this.

– The data are protein co-localisation, not signalling. Distances are nucleus centre to centre.

– Relapse ROIs have about 2.7× more blasts, which inflates every proximity measure. This is why the zone-composition and module results, which are measured at a fixed distance, carry the biology.

Files

spatial_proximity_final.py / .ipynb is the full analysis (identical code). **csv/** holds per-ROI values and every statistics table (P1–P5). **figures/** holds figs 1–12. **make_summary_pdf.py** builds this report.